

Zachary W. Ulissi

SENIOR RESEARCH SCIENTIST MANAGER · META FUNDAMENTAL AI RESEARCH

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Positions

Meta Fundamental AI Research (FAIR) SENIOR RESEARCH SCIENTIST MANAGER	San Francisco, CA (08/2024 - present)
Meta Fundamental AI Research (FAIR) RESEARCH SCIENTIST	San Francisco, CA (01/2023 - 08/2024)
Carnegie Mellon University ADJUNCT PROF. OF CHEM. ENG.	Pittsburgh, PA (01/2024 - present)
Carnegie Mellon University ASSOC. PROF. (W/O TENURE) OF CHEM. ENG. AND (COURTESY) MSE	Pittsburgh, PA (07/2022 - 12/2023)
Carnegie Mellon University ASST. PROF. OF CHEMICAL ENG. AND (COURTESY) MAT. SCI. AND ENG.	Pittsburgh, PA (08/2017 - 06/2022)
Stanford University / SLAC POSTDOCTORAL FELLOW, CHEMICAL ENGINEERING	Stanford, CA (07/2015 - 07/2017)

Education

Massachusetts Institute of Technology PH.D. IN CHEM. ENG.; M. STRANO AND R. BRAATZ	Cambridge, MA (09/2010 - 06/2015)
Churchill College, University of Cambridge M.A.ST. IN APPLIED MATH AND THEORETICAL PHYSICS	Cambridge, UK (09/2009 - 06/2010)
University of Delaware HONORS B.E. IN CHEMICAL ENGINEERING, HONORS B.S. IN PHYSICS	Newark, Delaware (09/2005 - 05/2009)

Honors & Awards

2022	Dean's Early Career Fellow , Outstanding research accomplishments and potential	Engineering, CMU
2021	G.T. Ladd Research Award , Outstanding research accomplishments and potential	Engineering, CMU
2020	AIChE 35 under 35 , Members under 35 who have made sig. contributions to the field of Chem. Eng.	AIChE
2020-2023	3M Non-Tenured Faculty Award , Research support and interaction with 3M	3M
2020-2022	ACS PRF Doctoral New Investigator , Research support for preliminary project	ACS PRF
2019-2023	Scott Institute for Energy Innovation Fellow , Logistics/support to promote energy research at CMU	CMU
2019	Scialog Fellow , Fellowship for the Scialog Conference in Energy Storage	Scialog Foundation
2018	Wimmer Fellow , Fellowship to improve 06-625 Chemical and Reactive Systems course	Eberly Center, CMU
2017	Team Science Award , "Materials disc., theory, and charact. of intermetallics for electro. CO2 reduction"	DOE EFRC Meeting
2010-2014	DOE CSGF Fellow , Four years of support, interaction with DOE computational science community	DOE
2009-2010	NSF GRFP Fellow , Three years of partial fellowship support	NSF

Teaching

UG: Numerical Methods and Machine Learning for Chemical Engineering Z ULISSI	06-325 (Fall 2022)
UG: Mathematical Methods of Chemical Engineering ZACHARY ULISSI	06-262 (Spring 2019-2022)
MS: Chemical and Reactive Systems ZACHARY ULISSI	06-625 (Fall 2017-2020)

Research Group (CMU)

Alumni

Nicholas Tiwari (PhD, Changed Advisor), Kirby Broderick (PhD, Changed Advisor), Unnatti Sharma (PhD, Changed Advisor), Joseph Musielewicz (PhD, Changed Advisor), Adeesh Kolluru (PhD, Changed Advisor), Xiaoxiao (Lory) Wang (PhD, Changed Advisor), Brook Wander (PhD, Changed Advisor), Rohan (Yuri) Sanspeur (PhD, Changed Advisor), Hilda Mera (MS 2023), Ethan Sunshine (PhD, Changed Advisor), Janghoon Ock (PhD, Changed Advisor), Javier Heras Domingo (Post-doc 2023, ICIQ), Tian Tian (Post-doc 2023, Faculty U.Alberta), Richard Tran (Post-doc 2023, Post-doc U.Houston), Kyung-Eun You (Post-doc 2023, Post-doc UPitt), Aaron Garrison (UG 2023, PhD MIT), Anna Maria Ferrante (UG 2024), Logan Meritz (UG 2025), Muhammed Shuaibi (PhD 2022, Meta FAIR), Kevin Tran (PhD 2021, Schrodinger), Amish Chovatiya (MS 2019, Intermolecular, now PhD Notre Dame), Katsuyuki Tomita (MS 2019, Nippon Steel), Nianhan (Kaylee) Tian (UG/MS 2020, PhD G.Tech), Seoin Back (PD, faculty at Sogang University), Zong Qian Yu (MS 2018), Wen Zhong (MS 2018, consulting), Hyukjae Kwark (BS 2020), Shiv Rekhi (MS 2019, co-advised w/ Kathryn Whitehead, PhD Lehigh), Tanmay Raj (MS 2019, Cargill), Qingyang Zhang (MS 2019, RA with Venkat Viswanathan), Sudheesh Ethirajan (MS 2020, PhD UC Davis), Jingxuan Li (MS 2020), Arundhati Madabhushi (MS 2020, Consultant at BCG), Matthew Adams (MS 2020, EPA), Rui Qi Chen (UG 2021, PhD GTech), Ketong Chen (UG 2022), Megan McGraw (UG 2022), Jasmine Fahrenkrug (UG, 2022), Saurabh Pattabiraman (MS 2020, PhD UC Davis), Ziheng (Daniel) Liu (UG 2021), Aini Palizhati (PhD 2022, Boston Consulting Group), Junwoong Yoon (PhD 2022, Meta), Rajesh Raju (Post-Doc, MS at MILA)

Invited Talks and Seminars

University of Notre Dame Dept of Chem. and Biom. Eng. ZACHARY ULISSI	South Bend, IN (9/2025)
Telluride Science ZACHARY ULISSI	Telluride, CO (07/2025)
SEMI Strategic Materials Conference Invited Talk ZACHARY ULISSI	San Jose, CA (6/2025)
3M Tech Talk ZACHARY ULISSI	Virtual (6/2025)
SUPrCat Seminar ZACHARY ULISSI	Virtual (6/2025)
Argonne National Lab Director's Colloquium ZACHARY ULISSI	Lemont, IL (5/2025)
UC Berkeley BIDMAP Seminar ZACHARY ULISSI	Berkeley, CA (5/2025)
ECOMATES School on in-silico Methods for Catalyst Design ZACHARY ULISSI	Dublin, IE (Remote) (3/2025)
Stanford CHE65Q Guest Lecture ZACHARY ULISSI	Stanford, CA (3/2025)
University of Chicago Department of Geophysical Sciences Seminar ZACHARY ULISSI	Chicago, IL (2/2025)
Learning on Graphs (LoG) Conference Keynote ZACHARY ULISSI	Remote (11/2024)
Block AI Keynote on AI for Science ZACHARY ULISSI	Oakland, CA (11/2024)
Northwestern University Research Seminar ZACHARY ULISSI	Evanston, IL (10/2024)
AIMED Workshop on Heterogeneous Catalysis ZACHARY ULISSI	Chicago, IL (10/2024)
UChicago Schmidt Science Fellows AI+Science Summer School ZACHARY ULISSI	Chicago, IL (7/2024)
Nobel Turing Challenge Initiative Keynote ZACHARY ULISSI	Singapore, SG (7/2024)
NTU University Research Seminar ZACHARY ULISSI	Nanyang, SG (7/2024)
Shell.ai Aryabhata Seminar Series ZACHARY ULISSI	US (Remote) (6/2024)
Simons Institute AI=Science Workshop ZACHARY ULISSI	Berkeley, CA (6/2024)
Stahl Forum ZACHARY ULISSI	Minneapolis, MN (4/2024)
Aionics Industrial Seminar ZACHARY ULISSI	US (Remote) (4/2024)
TACO (Taming Complexity Together) PhD Retreat ZACHARY ULISSI	Austria (Remote) (2/2024)
NYU/Simons Center for Comp. Phys. Chem. ZACHARY ULISSI	New York, NY (11/2023)
Northeastern University Mechanical and Industrial Eng. ZACHARY ULISSI	Boston, MA (11/2023)
LBL Advanced Light Source ZACHARY ULISSI	Berkeley (CA) (09/2023)
Climate Change AI Web Seminar ZACHARY ULISSI	US (Remote) (09/2023)
Foundations of Process Analytics and Machine Learning (FOPAM) ZACHARY ULISSI	Davis, CA (08/2023)
ACS Symposium on Data Science in Catalysis: Catalysis Informatics ZACHARY ULISSI	San Francisco, CA (08/2023)
ACS Symposium on Data Science in Catalysis ZACHARY ULISSI	San Francisco, CA (08/2023)
ACS Symposium in Honor of AJ Medford ZACHARY ULISSI	San Francisco, CA (08/2023)
Telluride Science ZACHARY ULISSI	Telluride, CO (07/2023)
Dow Chemical ZACHARY ULISSI	US (Remote) (07/2023)
ICLR ML4Materials Workshop ZACHARY ULISSI	US (Remote) (05/2023)
Georgia Tech Chemical and Biomolecular Engineering ZACHARY ULISSI	Atlanta, GA (4/2023)
Center for Environmentally Beneficial Catalysis ZACHARY ULISSI	US (remote) (3/2023)
Great Plains Catalysis Society ZACHARY ULISSI	US (remote) (11/2022)
Great Plains Catalysis Society ZACHARY ULISSI	US (remote) (11/2022)
Toyota Research Institute ZACHARY ULISSI	Los Altos, CA (remote) (11/2022)
Catalyst Modeling Symposium ZACHARY ULISSI	Rungstedgaard, DK (9/2022)
ACS Fall National Meeting ZACHARY ULISSI	Chicago, IL (8/2022)
NIST Artificial Intelligence for Materials Science (AIMS) Symposium ZACHARY ULISSI	Gaithersburg, MD (remote) (7/2022)
Nano Korea ZACHARY ULISSI	Korea (remote) (7/2022)
Machine Learning for Materials (ML4M) ZACHARY ULISSI	Trieste, Italy (5/2022)
NanoGe Conference ZACHARY ULISSI	Spain (remote) (3/2022)
Monash University ZACHARY ULISSI	Melbourne, Australia (remote) (3/2022)
Heraeus Workshop ZACHARY ULISSI	Germany (remote) (1/2022)
Technical University Delft ZACHARY ULISSI	Delft, NL (remote) (11/2021)
3M Young Faculty Symposium ZACHARY ULISSI	St Paul, MN (remote) (11/2021)
German Young Professional Engineers Seminar Series (NaWuReT) ZACHARY ULISSI	Germany (remote) (10/2021)
ACS Fall Meeting Symposium Keynote ZACHARY ULISSI	Atlanta, GA (8/2021)
Materials Project Seminar Series ZACHARY ULISSI	Berkeley, CA (remote) (8/2021)
Statistical Thermodynamics and Molecular Simulation (STMS) Series ZACHARY ULISSI	Remote (6/2021)
Telluride Workshop on Computational Materials Chemistry ZACHARY ULISSI	Telluride, CO (6/2021)
Colloid and Surface science Symposium ZACHARY ULISSI	State College, PA (remote) (6/2021)
International Workshop on Computer-Aided Materials Discovery ZACHARY ULISSI	Korea (remote) (6/2021)
OPTIMADE workshop ZACHARY ULISSI	Louvain, Belgium (remote) (6/2021)
DOE NERSC Data Seminar ZACHARY ULISSI	Berkeley, CA (remote) (4/2021)

North Carolina State University Dept of Chem. and Biom. Eng.	ZACHARY ULISSI	Raleigh, NC (remote)	(4/2021)
Penn Institute for Computational Science	ZACHARY ULISSI	Philadelphia, PA (remote)	(3/2021)
3M Young Faculty Symposium	ZACHARY ULISSI	St Paul, MN (remote)	(2/2021)
Matminer Singapore Seminar Series	ZACHARY ULISSI	Singapore (remote)	(12/2020)
Fritz-Haber Institute Theory Seminar	ZACHARY ULISSI	Berlin, GE (remote)	(12/2020)
DTRA Annual Review External Speaker	ZACHARY ULISSI	Aberdeen, MD (remote)	(11/2020)
3M	ZACHARY ULISSI	St Paul, MN (remote)	(11/2020)
Dow	ZACHARY ULISSI	Midlands, MI (remote)	(11/2020)
EcoCAT Symposium	ZACHARY ULISSI	Barcelona, Spain (remote)	(11/2020)
NERSC Deep Learning for Science Lecture	ZACHARY ULISSI	LBL, Berkeley, CA (remote)	(10/2020)
RAPID ML in Sci/Eng. Seminar	ZACHARY ULISSI	Newark, DE (remote)	(7/2020)
DTU Catalysis Theory Seminar	ZACHARY ULISSI	Lyngby, DK (remote)	(6/2020)
Lyondell Bassell	ZACHARY ULISSI	Houston, TX (remote)	(5/2020)
Citrine Informatics	ZACHARY ULISSI	Redwood City, CA	(3/2020)
DOE NERSC Data Seminar	ZACHARY ULISSI	LBL, Berkeley, CA	(3/2020)
Symposium on AI in Chemical Engineering	ZACHARY ULISSI	ETH Zürich, Switzerland	(2/2020)
MRS Fall Meeting	ZACHARY ULISSI	Boston, MA	(12/2019)
Chemistry Meets AI Symposium	ZACHARY ULISSI	KAIST, Daejeon, Korea	(11/2019)
International Conference on Molecular Simulation	ZACHARY ULISSI	Jeju Island, Korea	(11/2019)
University of Delaware Department of Physics	ZACHARY ULISSI	Newark, DE	(10/2019)
University of Washington Department of Chemical Engineering	ZACHARY ULISSI	Seattle, WA	(10/2019)
NIST Artificial Intelligence for Materials Science (AIMS) Workshop	ZACHARY ULISSI	Gaithersburg, MD	(08/2019)
Lehigh University Inst for Data, Intelligent Systems, and Comp	Z ULISSI	Bethlehem, PA	(05/2019)
Keynote for Virginia Tech Chem. Eng. Graduate Student Symposium	ZACHARY ULISSI	Blacksburg, VA	(04/2019)
ExxonMobil Research and Development	ZACHARY ULISSI	Annandale, PA	(04/2019)
COMET NSF IGERT Training Center Seminar	ZACHARY ULISSI	State College, PA	(11/2018)
Pittsburgh Cleveland Catalysis Society Annual Symposium	ZACHARY ULISSI	Pittsburgh, PA	(6/2018)
Workshop: Accelerating the Dev. of Energy Mat. and Novel Structural Mat.	Z ULISSI	Toronto, CA	(5/2018)
Materials Science Division, Lawrence Livermore National Laboratory	ZACHARY ULISSI	Livermore, CA	(5/2018)
ACS Spring Meeting	ZACHARY ULISSI	New Orleans, LA	(3/2018)
University of Auckland Department of Chem. Engineering	ZACHARY ULISSI	Auckland, NZ	(2/2018)
University of Toronto Department of Electrical Engineering	ZACHARY ULISSI	Toronto, CA	(11/2017)
Institute for Pure and Applied Mathematics Workshop Series	ZACHARY ULISSI	Los Angeles, CA	(10/2017)
Machine Learning for Energy Materials CIFAR Workshop	ZACHARY ULISSI	Toronto, CA	(5/2017)

Publications, Pre-Prints, Conf. Papers, Chapters, and Tech. Reports _____

[94]Open Molecular Crystals 2025 (OMC25) Dataset and Models

V. Gharakhanyan, L. Barroso-Luque, Y. Yang, M. Shuaibi, K. Michel, D. S. Levine, M. Dzamba, X. Fu, M. Gao, X. Liu, H. Ni, K. Noori, B. M. Wood, M. Uyttendaele, A. Boromand, C. L. Zitnick, N. Marom, Z. W. Ulissi, A. Sriram *arXiv preprint arxiv:2508.02651*. 2025

[93]FastCSP: Accelerated Molecular Crystal Structure Prediction with Universal Model for Atoms

V. Gharakhanyan, Y. Yang, L. Barroso-Luque, M. Shuaibi, D. S. Levine, K. Michel, V. Bernat, M. Dzamba, X. Fu, M. Gao, X. Liu, K. Noori, L. J. Purvis, T. Rao, B. M. Wood, A. Rizvi, M. Uyttendaele, A. J. Ouderkirk, C. Daraio, C. L. Zitnick, A. Boromand, N. Marom, Z. W. Ulissi, A. Sriram *arXiv preprint arxiv:2508.02641*. 2025

[92]Multi-Physics Inverse Design of Varifocal Optical Devices using Data-Driven Surrogates and Differential Modeling

Z. Jin, Z. Liu, N. Elabbasi, Z. Ulissi, G. X. Gu, Z. Nie *arXiv preprint arxiv:2503.18911*. 2025

[91]All-atom Diffusion Transformers: Unified generative modelling of molecules and materials

C. K. Joshi, X. Fu, Y.-L. Liao, V. Gharakhanyan, B. K. Miller, A. Sriram, Z. W. Ulissi *arXiv preprint arxiv:2503.03965*. 2025

[90]The Open Molecules 2025 (OMol25) Dataset, Evaluations, and Models

D. S. Levine, M. Shuaibi, E. W. C. Spotte-Smith, M. G. Taylor, M. R. Hasyim, K. Michel, I. Batatia, G. Csányi, M. Dzamba, P. Eastman, N. C. Frey, X. Fu, V. Gharakhanyan, A. S. Krishnapriyan, J. A. Rackers, S. Raja, A. Rizvi, A. S. Rosen, Z. Ulissi, S. Vargas, C. L. Zitnick, S. M. Blau, B. M. Wood *arXiv preprint arxiv:2505.08762*. 2025

[89]The Open Catalyst 2025 (OC25) Dataset and Models for Solid-Liquid Interfaces

S. J. Sahoo, M. Maraschin, D. S. Levine, Z. Ulissi, C. L. Zitnick, J. B. Varley, J. A. Gauthier, N. Govindarajan, M. Shuaibi *arXiv preprint arxiv:2509.17862*. 2025

[88]The Open DAC 2025 Dataset for Sorbent Discovery in Direct Air Capture

A. Sriram, L. M. Brabson, X. Yu, S. Choi, K. Abdelmaqsoud, E. Moubarak, P. Haan, S. Löwe, J. Brehmer, J. R. Kitchin, M. Welling, C. L. Zitnick, Z. Ulissi, A. J. Medford, D. S. Sholl *arXiv preprint arxiv:2508.03162*. 2025

- [87] **CatTSunami: Accelerating Transition State Energy Calculations with Pretrained Graph Neural Networks**
B. Wander, M. Shuaibi, J. R. Kitchin, Z. W. Ulissi, C. L. Zitnick *ACS Catalysis* pp. 5283–5294, 2025
- [86] **Constraint Active Search in Process Window Optimization for Powder Feed Directed Energy Deposition**
X. Wang, J. A. Loli, Z. W. Ulissi, M. P. Boer, B. A. Webler, R. C. Kurchin *Integrating Materials and Manufacturing Innovation* pp. 106–114, 2025
- [85] **UMA: A Family of Universal Models for Atoms**
B. M. Wood, M. Dzamba, X. Fu, M. Gao, M. Shuaibi, L. Barroso-Luque, K. Abdelmaqsoud, V. Gharakhanyan, J. R. Kitchin, D. S. Levine, K. Michel, A. Sriram, T. Cohen, A. Das, A. Rizvi, S. J. Sahoo, Z. W. Ulissi, C. L. Zitnick *arXiv preprint arXiv:2506.23971*. 2025
- [84] **Roadmap for Transforming Heterogeneous Catalysis with Artificial Intelligence**
H. Xin, J. Kitchin, N. López, N. Schweitzer, N. Artrith, F. Che, L. Grabow, K. Gunasooriya, H. Kulik, T. Laino *ChemRxiv*. 2025
- [83] **Genetic Algorithm-Accelerated Computational Discovery of Liquid Crystal Polymers with Enhanced Optical Properties**
J. Zhou, Y. Huang, A. Boromand, K. Noori, L. Purvis, C. Oh, L. Lu, Z. W. Ulissi, V. Gharakhanyan, X. Zhang *arXiv preprint arXiv:2505.13477*. 2025
- [82] **Pourbaix Machine Learning Framework Identifies Acidic Water Oxidation Catalysts Exhibiting Suppressed Ruthenium Dissolution**
J. Abed, J. Heras-Domingo, R. Y. Sanspeur, M. Luo, W. Alnoush, D. M. Meira, H. Wang, J. Wang, J. Zhou, D. Zhou *Journal of the American Chemical Society*. 2024
- [81] **Open Catalyst Experiments 2024 (OCx24): Bridging Experiments and Computational Models**
J. Abed, J. Kim, M. Shuaibi, B. Wander, B. Duijf, S. Mahesh, H. Lee, V. Gharakhanyan, S. Hoogland, E. Irtem, J. Lan, N. Schouten, A. U. Vijayakumar, J. Hatrick-Simpers, J. R. Kitchin, Z. W. Ulissi, A. Vugt, E. H. Sargent, D. Sinton, C. L. Zitnick . 2024
- [80] **Open materials 2024 (omat24) inorganic materials dataset and models**
L. Barroso-Luque, M. Shuaibi, X. Fu, B. M. Wood, M. Dzamba, M. Gao, A. Rizvi, C. L. Zitnick, Z. W. Ulissi *arXiv preprint arXiv:2410.12771*. 2024
- [79] **Adapting OC20-Trained EquiformerV2 Models for High-Entropy Materials**
C. M. Clausen, J. Rossmeisl, Z. W. Ulissi *The Journal of Physical Chemistry C* pp. 11190–11195, 2024
- [78] **Fine-Tuned Language Models Generate Stable Inorganic Materials as Text**
N. Gruver, A. Sriram, A. Madotto, A. Wilson, C. L. Zitnick, Z. Ulissi *ICLR*. 2024
- [77] **Enumeration of surface site nuclearity and shape in a database of intermetallic low-index surface facets**
U. Sharma, A. Nguyen, J. R. Kitchin, Z. W. Ulissi, M. J. Janik *Journal of Catalysis* p. 115795, 2024
- [76] **From Molecules to Materials: Pre-training Large Generalizable Models for Atomic Property Prediction**
N. Shoghi, A. Kolluru, J. R. Kitchin, Z. W. Ulissi, C. L. Zitnick, B. M. Wood *ICLR*. 2024
- [75] **The Open DAC 2023 Dataset and Challenges for Sorbent Discovery in Direct Air Capture**
A. Sriram, S. Choi, X. Yu, L. M. Brabson, A. Das, Z. Ulissi, M. Uyttendaele, A. J. Medford, D. S. Sholl *ACS Central Science* pp. 923–941, 2024
- [74] **chap. Chapter 6: Practical Application of Machine Learning in Catalysis** Z. W. ULISSI, K. TRAN, J. YOON, M. SHUAIBI, M. LIU, N. ZHAN, K. BRODERICK, J. R. KITCHIN “COMPUTATIONAL CATALYSIS”, 2024
- [73] **Generalization of graph-based active learning relaxation strategies across materials**
X. Wang, J. Musielewicz, R. Tran, S. K. Ethirajan, X. Fu, H. Mera, J. R. Kitchin, R. C. Kurchin, Z. W. Ulissi *Machine Learning: Science and Technology* p. 025018, 2024
- [72] **Applying Large Graph Neural Networks to Predict Transition Metal Complex Energies Using the tmQM_wB97MV Data Set**
A. G. Garrison, J. Heras-Domingo, J. R. Kitchin, G. Passos Gomes, Z. W. Ulissi, S. M. Blau *Journal of Chemical Information and Modeling*. 2023
- [71] **AdsorbML: a leap in efficiency for adsorption energy calculations using generalizable machine learning potentials**
J. Lan, A. Palizhati, M. Shuaibi, B. M. Wood, B. Wander, A. Das, M. Uyttendaele, C. L. Zitnick, Z. W. Ulissi *npj Computational Materials* p. 172, 2023
- [70] **Beyond independent error assumptions in large GNN atomistic models**
J. Ock, T. Tian, J. Kitchin, Z. Ulissi *The Journal of Chemical Physics*. 2023
- [69] **Cluster-MLP: An Active Learning Genetic Algorithm Framework for Accelerated Discovery of Global Minimum Configurations of Pure and Alloyed Nanoclusters**
R. K. Raju, S. Sivakumar, X. Wang, Z. W. Ulissi *Journal of Chemical Information and Modeling*. 2023
- [68] **WhereWulff: A Semiautonomous Workflow for Systematic Catalyst Surface Reactivity under Reaction Conditions**
R. Y. Sanspeur, J. Heras-Domingo, J. R. Kitchin, Z. Ulissi *Journal of Chemical Information and Modeling* pp. 2427–2437, 2023
- [67] **AmpTorch: A Python package for scalable fingerprint-based neural network training on multi-element systems with integrated uncertainty quantification**

M. Shuaibi, Y. Hu, X. Lei, B. M. Comer, M. Adams, J. Paras, R. Q. Chen, E. Musa, J. Musielewicz, A. A. Peterson *Journal of Open Source Software* p. 5035, 2023

[66]Chemical Properties from Graph Neural Network-Predicted Electron Densities

E. M. Sunshine, M. Shuaibi, Z. W. Ulissi, J. R. Kitchin *The Journal of Physical Chemistry C* pp. 23459–23466, 2023

[65]The Open Catalyst 2022 (OC22) Dataset and Challenges for Oxide Electrocatalysts

R. Tran, J. Lan, M. Shuaibi, B. M. Wood, S. Goyal, A. Das, J. Heras-Domingo, A. Kolluru, A. Rizvi, N. Shoghi, A. Sriram, F. Therrien, J. Abed, O. Voznyy, E. H. Sargent, Z. Ulissi, C. L. Zitnick *ACS Catalysis* pp. 3066–3084, 2023

[64]Identifying limitations in screening high-throughput photocatalytic bimetallic nanoparticles with machine-learned hydrogen adsorptions

K. Broderick, E. Lopato, B. Wander, S. Bernhard, J. Kitchin, Z. Ulissi *Applied Catalysis B: Environmental* p. 121959, 2023

[63]How Do Graph Networks Generalize to Large and Diverse Molecular Systems?

J. Gasteiger, M. Shuaibi, A. Sriram, S. Günnemann, Z. Ulissi, C. L. Zitnick, A. Das *arXiv preprint arXiv:2204.02782*. 2022

[62]Transfer learning using attentions across atomic systems with graph neural networks (TAAG)

A. Kolluru, N. Shoghi, M. Shuaibi, S. Goyal, A. Das, C. L. Zitnick, Z. Ulissi *The Journal of Chemical Physics* p. 184702, 2022

[61]Detailed Microkinetics for the Oxidation of Exhaust Gas Emissions through Automated Mechanism Generation

B. Kreitz, P. Lott, J. Bae, K. Blöndal, S. Angeli, Z. W. Ulissi, F. Studt, C. F. Goldsmith, O. Deutschmann *ACS Catalysis* pp. 11137–11151, 2022

[60]Predicting Oxidation Behavior of Multi-Principal Element Alloys by Machine Learning Methods

J. A. Loli, A. R. Chovatiya, Y. He, Z. W. Ulissi, M. P. Boer, B. A. Webler *Oxidation of Metals* pp. 429–450, 2022

[59]Heterogeneous Catalysis in Grammar School

J. T. Margraf, Z. W. Ulissi, Y. Jung, K. Reuter *The Journal of Physical Chemistry C* pp. 2931–2936, 2022

[58]Site Geometry as a Descriptor for Catalyst Selectivity in Intermetallics

U. Sharma, A. Nguyen, M. J. Janik, Z. Ulissi *Preprint available at SSRN 4145497*. 2022

[57]Screening of bimetallic electrocatalysts for water purification with machine learning

R. Tran, D. Wang, R. Kingsbury, A. Palizhati, K. A. Persson, A. Jain, Z. W. Ulissi *The Journal of Chemical Physics* p. 074102, 2022

[56]The Open Catalyst Challenge 2021: Competition Report

A. Das, M. Shuaibi, A. Palizhati, S. Goyal, A. Grover, A. Kolluru, J. Lan, A. Rizvi, A. Sriram, B. Wood, D. Parikh, Z. Ulissi, C. L. Zitnick, G. Ke, S. Zheng, Y. Shi, D. He, T.-Y. Liu, C. Ying, J. You, Y. He, R. Grigoriev, R. Lukin, A. Yarullin, M. Faleev pp. 29–40, 2022

[55]Robust and scalable uncertainty estimation with conformal prediction for machine-learned interatomic potentials

Y. Hu, J. Musielewicz, Z. W. Ulissi, A. J. Medford *Machine Learning: Science and Technology* p. 045028, 2022

[54]Spherical Channels for Modeling Atomic Interactions

C. L. Zitnick, A. Das, A. Kolluru, J. Lan, M. Shuaibi, A. Sriram, Z. Ulissi, B. Wood *NeurIPS*. 2022

[53]FINETUNA: Fine-tuning Accelerated Molecular Simulations

J. Musielewicz, X. Wang, T. Tian, Z. W. Ulissi *Machine Learning: Science and Technology*. 2022

[52]Catlas: an automated framework for catalyst discovery demonstrated for direct syngas conversion

B. Wander, K. Broderick, Z. W. Ulissi *Catal. Sci. Technol.* pp. -, 2022

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External Support (CMU)

Toyota Research Institute	PI: ZACHARY ULISSI, CO-PI: JEFFREY SCHNEIDER	(2021-2022)
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DOE NERSC Cori Supercomputer Allocation	PI: ZACHARY ULISSI	28M SU (2021)
DOE NERSC Cori Supercomputer Allocation	PI: ZACHARY ULISSI	28M SU (2020)
DOE NERSC Cori Supercomputer Allocation	PI: ZACHARY ULISSI	16M SU (2019)
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Service and Committees (CMU)

CMU	Facilities and Space (2019-), Graduate Recruitment (2018-2019), AIChE Undergraduate Adviser (2018-), Chemical Engineering Cube (ChemE Cube) advisor (2020-), Undergraduate Committee (2019-), CIT Cluster Administration (2017-), CIT ad-hoc Research Computing Committee (2020-)
Reviewer	Nature Communications, ACS Catalysis, Journal of Physical Chemistry Letters, Journal of Physical Chemistry C, Journal of Vacuum Science and Technology, among many others
Symposia	AIChE Annual Meeting (2018-present), MRS Fall Meeting (2019), CMU/GT Machine Learning in Science in Engineering (2018), ACS Spring Meeting (2018)